Q.26 Define measurement. Explain direct and indirect measurement.

Q.27 Compare attraction and repulsion type moving iron instruments.

Q.28 What are the various factors on which PH value depends?

Q.29 Write a short note on thermocouple and thermopiles.

Q.30 Differentiate between thermistors and RTDs.

Q.31 Give salient features of LVDT.

Q.32 Explain the working principle of piezoelectric transducers.

Q.33 Why ammeter is connected in series and voltmeter is connected in parallel.

Q.34 Why eddy current damping is the most efficient damping method.

Q.35 Define gauge factor. Give uses of electrical strain gauge.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the working, construction and application of LCR meters using suitable diagrams.

Q.37 Explain the construction and working of CRO with the help of a diagram.

Q.38 Explain different types of errors in the measuring instruments. How these errors can be minimized.





Roll No. .................

**3rd Semester/**
**Elect., Power Stat. Engg., Elect & Eltx. Engg.**
**Subject: Electrical Measurements and Measuring Instruments**

Time : 3 Hrs. M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Series electro magnet in induction type energy meter consists of _____.

a) L Shaped laminations b) E Shaped laminations

c) T Shaped laminations d) U Shaped laminations

Q.2 The household energy meter is ___.

a) An indicating instrument

b) A recording instrument

c) An integrating instrument

d) None of these

Q.3 The supply to the megger is given by ___.

a) AC motor

b) AC generator

c) Permanent magnet DC generator

d) DC Motor

Q.4 Capacitive transducer is used to measure

a) Displacement b) Thickness

c) Level d) All of these

Q.5 Input impedance of C.R.O. is

a) Zero b) Around 100 ohms

c) Around 1000 ohms d) Around one mega ohms

Q.6 Air is a mixture of

a) Dry air b) Water vapour

c) Both d) None of the above

Q.7 Force is defined as

a) Velocity x displacement

b) Velocity x acceleration

c) Mass x displacement

d) Mass x acceleration

Q.8 Aquadag is used in CRO to collect

a) Primary electrons

b) Secondary emission electrons

c) Both primary & secondary emission

d) None of the above

Q.9 Potentiometer is a _______ transducer

a) Displacement b) Force

c) Torque d) None of these

Q.10 Pressure is measured in

a) Newton b) Pascal

c) Weber d) Tesla



## SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 LVDT is an inductive transducer (True/False)

Q.12 The closeness of the measured value to the true value is known as precision. (True/False)

Q.13 Thermoelectric transducers use _______ effect.

Q.14 Scale of a moving iron instrument is __________.

Q.15 Resistance temperature detectors are made up of _____.

Q.16 Expand LVDT.

Q.17 What is an error?

Q.18 Explain the use of controlling torque.

Q.19 Give an application of clamp-on meter.

Q.20 Define pyrometry.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Give principle and working of strain gauge.

Q.22 Differentiate between active and passive transducers.

Q.23 Explain the construction and working of a potential transformer.

Q.24 What is the basic working principle of PMMC meters?

Q.25 Explain the operating principle of earth tester for measuring earth resistance.